



AZ-900

Microsoft Azure Fundamentals

STUDY NOTES

Complete Azure Fundamentals Reference Guide

2025 EDITION

10

Sections

40+

Topics

200+

Key Points

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Microsoft Azure Certification Preparation Guide

AZ-900 Study Notes | 2025 Edition

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10 Sections

40+ Topics

200+ Key Points

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PREFACE & INTRODUCTION

Welcome to the AZ-900 Microsoft Azure Fundamentals Study Notes — your concise, structured reference guide for rapid revision and quick concept lookup before the certification exam.

Unlike the Q&A format, this guide presents each topic with key concepts, important points to remember, and a quick reference table of key terms, comparisons, and differences — making it ideal for last-minute revision or building mental models of Azure services.

How to use this guide: Each section covers a major AZ-900 exam domain. Under each topic heading you will find bullet-point key concepts followed by a quick reference table. Pay special attention to bold terms — these are frequently tested on the exam.

Topics covered: Cloud Concepts • Azure Architecture • Compute Services • Storage Services • Networking • Identity & Security • Management & Governance • Databases & AI • DevOps & Developer Tools • Pricing, Support & Exam Tips

Remember: AZ-900 tests CONCEPTS, not implementation. Focus on understanding what each service does and when to use it. Practice on Microsoft Learn's free AZ-900 learning path!

— Akhilesh Chaudhari

Section 1: Cloud Concepts

Cloud Models, Benefits, Service Types & Shared Responsibility

What is Cloud Computing?

- Delivery of computing services (servers, storage, databases, networking, software) over the internet
- Pay only for what you use — no upfront hardware investment required
- CapEx (Capital Expenditure): upfront spend on hardware — traditional on-premises model
- OpEx (Operational Expenditure): ongoing pay-as-you-go spending — cloud model
- Cloud shifts IT from CapEx to OpEx, reducing large upfront investments

CapEx model	Buy servers, pay upfront, own assets, depreciation over time
OpEx model	Pay monthly for cloud services, scale up/down anytime
Consumption-based	Pay only for resources used — no waste from idle capacity
Economies of scale	Microsoft buys at huge scale □ passes savings to customers

Cloud Deployment Models

- Public Cloud: owned by Microsoft, shared multi-tenant infrastructure, accessed over internet — lowest cost
- Private Cloud: dedicated to one organization, on-premises or hosted, greater control, higher cost
- Hybrid Cloud: combines public + private — data stays on-prem for compliance, burst to public for scale
- Multi-Cloud: using multiple cloud providers (Azure + AWS) — avoid vendor lock-in
- Azure Arc enables managing on-premises and multi-cloud resources through Azure

Public Cloud	Shared, internet-accessible, lowest cost, managed by Microsoft
Private Cloud	Dedicated, on-premises or hosted, full control, highest cost
Hybrid Cloud	Best of both — compliance data on-prem, scale in cloud
Azure Stack	Run Azure services on-premises for disconnected/compliant scenarios

Cloud Service Models (IaaS / PaaS / SaaS)

- IaaS (Infrastructure as a Service): you manage OS, apps, data — Azure provides VMs, storage, network
- PaaS (Platform as a Service): you manage apps and data — Azure manages OS, runtime, middleware
- SaaS (Software as a Service): you manage data only — Azure/vendor manages everything else
- IaaS examples: Azure VMs, Azure Disk Storage. PaaS: App Service, Azure SQL DB. SaaS: Microsoft 365, Dynamics
- Moving from IaaS □ SaaS increases Microsoft's responsibility and reduces yours

IaaS	VM, storage, network – YOU manage OS and above
PaaS	App Service, Azure SQL – YOU manage app code and data only
SaaS	Microsoft 365, Teams – YOU manage only data and user access
Shared Responsibility	Responsibility shifts from customer to Microsoft as you move IaaS→PaaS→SaaS

Benefits of Cloud Computing

- High Availability: SLAs guarantee uptime (99.9% = 8.7 hrs/year downtime; 99.99% = 52 min/year)
- Scalability: vertical (add CPU/RAM to existing) + horizontal (add more instances)
- Elasticity: automatic dynamic scaling based on real-time demand – no manual intervention
- Reliability: distributed design, geo-redundancy, automatic failover
- Security: enterprise-grade, Microsoft manages physical security, encryption, compliance
- Governance: Azure Policy, RBAC, Blueprints ensure resources meet organizational standards
- Manageability: Portal, CLI, PowerShell, REST API, ARM templates for automation

High Availability	99.9% SLA = 8.7 hrs downtime/year 99.99% = 52 min/year
Scalability	Vertical: bigger VM Horizontal: more VMs
Elasticity	Auto-scale in/out based on demand metrics (CPU, requests)
Disaster Recovery	Azure Site Recovery, geo-redundant storage, backup services

Section 2: Azure Architecture

Regions, Availability Zones, Region Pairs & Global Infrastructure

Azure Global Infrastructure

- Geography: discrete market (country/group of countries) preserving data residency — e.g., United States, Europe
- Region: set of datacenters connected by low-latency network — 60+ regions worldwide (most of any cloud)
- Availability Zone (AZ): physically separate datacenter within a region — independent power, cooling, networking
- Deploying across AZs provides 99.99% SLA — protects against whole datacenter failure
- Region Pair: two regions 300+ miles apart in same geography — used for disaster recovery

Geography	<i>Data residency boundary — e.g., EU, US, Asia Pacific</i>
Region	<i>East US, West Europe, Southeast Asia — where resources deploy</i>
Availability Zone	<i>3 separate DCs per zone-enabled region — for 99.99% SLA</i>
Region Pair	<i>East US ↔ West US UK South ↔ UK West — DR partner</i>
Availability Set	<i>VMs spread across fault/update domains within ONE datacenter — 99.95% SLA</i>

Azure Management Hierarchy

- Management Groups: top level — organize subscriptions, apply governance across multiple subs (up to 6 levels deep)
- Subscriptions: billing unit and policy boundary — each resource belongs to exactly one subscription
- Resource Groups: logical containers for related resources — same lifecycle (deploy/delete together)
- Resources: individual services (VM, database, storage account) — must be in one RG, one region
- Policies and RBAC at higher levels are inherited by all lower levels

Management Group	<i>Governs multiple subscriptions — apply Policy/RBAC once for all</i>
Subscription	<i>Billing unit, quota boundary — one invoice per subscription</i>
Resource Group	<i>Logical container — organize by app, env, or team</i>
Resource	<i>Actual service instance — VM, SQL DB, Storage Account, etc.</i>
Azure Resource Manager	<i>All operations go through ARM — consistent management layer</i>

Azure Resource Manager (ARM)

- ARM is the deployment and management service for Azure — all operations (Portal, CLI, API) go through ARM
- ARM Templates: JSON files defining infrastructure as code — declarative, idempotent, version-controllable

- Bicep: cleaner DSL that compiles to ARM JSON – preferred for Azure-native IaC
- Tags: name-value pairs on resources for organization, billing, automation – up to 50 tags per resource
- Tags enable cost tracking by department, environment, or project (e.g., Environment:Prod, CostCenter:Finance)

ARM Template	<i>JSON declarative IaC – consistent, repeatable deployments</i>
Bicep	<i>Cleaner ARM alternative – compiles to ARM JSON, Azure-native</i>
Terraform	<i>Multi-cloud IaC (Azure+AWS+GCP) – uses AzureRM provider</i>
Azure Tags	<i>key:value metadata – filter billing, automate, enforce governance</i>

Section 3: Azure Compute Services

VMs, App Service, Containers, Functions & Serverless

Virtual Machines & Scale Sets

- Azure VMs: IaaS — full control of OS, apps, configuration. Windows or Linux. Pay per second
- VM Scale Sets (VMSS): auto-scale group of identical VMs — scale out on demand, scale in to save cost
- Availability Set: spread VMs across fault domains (rack) + update domains in ONE datacenter — 99.95% SLA
- Availability Zones: spread VMs across separate DCs — 99.99% SLA (preferred for new deployments)
- Pricing options: Pay-as-you-go | Reserved (1-3 yr, 72% saving) | Spot (90% saving, evictable) | Hybrid Benefit

Pay-as-you-go	No commitment, highest flexibility, billed per second
Reserved Instances	1 or 3 year commitment — up to 72% savings vs PAYG
Spot VMs	Unused Azure capacity — up to 90% discount, can be evicted anytime
Azure Hybrid Benefit	Use existing Windows/SQL Server licenses — save up to 85%
Azure Dedicated Host	Physical server for one subscription — compliance/isolation

Azure App Service & Containers

- App Service: fully managed PaaS for web apps, APIs, mobile backends — supports .NET, Java, Python, Node.js, PHP
- Deployment Slots: staging/production slots — zero-downtime swap to production, instant rollback
- Azure Container Instances (ACI): fastest way to run containers — no VM management, pay per second
- Azure Kubernetes Service (AKS): managed Kubernetes — Microsoft manages control plane, you manage workloads
- Azure Container Registry (ACR): private Docker registry — store, scan, replicate container images

App Service	PaaS web hosting — OS patching and scaling handled by Azure
Deployment Slots	Blue-green deploy — swap staging to production with zero downtime
ACI	Run single container quickly — no cluster needed, serverless containers
AKS	Managed Kubernetes — Microsoft manages master nodes, you deploy apps
ACR	Private container registry — build, store, scan, replicate images

Serverless & Functions

- Azure Functions: run small code snippets triggered by events — no server management, pay per execution
- Triggers: HTTP request, Timer schedule, Queue message, Blob upload, Event Hub, Cosmos DB change
- Consumption Plan: auto-scale to zero, pay per execution, cold start latency — best for infrequent workloads
- Premium Plan: pre-warmed (no cold start), VNet integration, extended timeout — best for production
- Logic Apps: no-code workflow automation — 200+ connectors (Office 365, Salesforce, SAP, Twitter)

Azure Functions	<i>Event-driven serverless code — runs and scales automatically</i>
Consumption Plan	<i>Pay per execution — scales to zero when idle (cold starts)</i>
Premium Plan	<i>Always warm, VNet support — for production serverless workloads</i>
Logic Apps	<i>Visual workflow builder — no-code integration between services</i>
Azure Virtual Desktop	<i>Cloud-based Windows desktop — any device, remote work</i>

Section 4: Azure Storage Services

Blob, Files, Queues, Disks, Redundancy & Data Transfer

Azure Storage Services Overview

- Blob Storage: unstructured data — images, videos, backups, logs. Block/Append/Page blobs
- Azure Files: managed SMB/NFS file shares — mountable by Windows, Linux, macOS
- Queue Storage: messages up to 64 KB, millions in queue — async decoupled communication
- Table Storage: NoSQL key-value store — fast, cheap, structured non-relational data
- Azure Disk Storage: persistent block volumes for VMs — Ultra/Premium SSD/Standard SSD/HDD

Blob Storage	Object store for any unstructured data — accessed via REST/SDK
Azure Files	SMB/NFS cloud file share — mount as network drive on any OS
Queue Storage	Decouple producers/consumers — messages retained up to 7 days
Azure Disk	VM persistent disks — managed disks recommended for all new VMs
Azure CDN	Cache static content at edge nodes near users — reduce latency

Blob Access Tiers & Lifecycle

- Hot: frequently accessed — highest storage cost, lowest access cost
- Cool: infrequent access (30+ days) — lower storage cost, higher access cost
- Cold: rare access (90+ days) — even lower storage cost
- Archive: rarely accessed (180+ days) — lowest cost, data is OFFLINE, rehydrate takes hours
- Lifecycle Management: policies auto-move blobs between tiers or delete them based on age

Hot tier	Frequent access — high storage cost, low access cost
Cool tier	30+ day retention — balanced cost for infrequent data
Archive tier	Lowest cost, offline — rehydrate to Hot/Cool before access
Lifecycle policy	Auto-tier: Hot → Cool after 30d, → Archive after 180d, delete after 365d
Soft delete	Protects against accidental deletion — retain for 1-365 days

Storage Redundancy Options

- LRS (Locally Redundant): 3 copies in 1 datacenter — cheapest, no geo protection
- ZRS (Zone-Redundant): 3 copies across 3 AZs in 1 region — datacenter failure protection
- GRS (Geo-Redundant): LRS in primary + 3 copies in secondary region — regional disaster protection
- GZRS (Geo-Zone-Redundant): ZRS in primary + 3 copies in secondary — highest durability (16 nines)
- Read-access variants: RA-GRS and RA-GZRS — allow reading from secondary region

LRS	3 copies, 1 DC — cheapest, no regional protection
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ZRS	3 copies, 3 AZs – survives single datacenter failure
GRS	6 copies, 2 regions – survives entire region failure
GZRS	ZRS primary + GRS secondary – maximum durability (16 nines)
Azure Data Box	Physical device for large data migration when network too slow

Section 5: Azure Networking

VNet, NSG, VPN, ExpressRoute, Load Balancer & DNS

Virtual Networks & Security

- VNet: private network in Azure — define address space (CIDR), divided into subnets
- NSG (Network Security Group): allow/deny rules by IP, port, protocol — applied to subnet or NIC
- NSG rules evaluated by priority (lower number = higher priority) — first match wins
- Azure Firewall: managed stateful FQDN-based firewall — centralized, scalable, threat intelligence
- VNet Peering: direct Microsoft backbone connection between VNets — non-transitive, no gateway needed

VNet	Private network — address space e.g., 10.0.0.0/16, divided into subnets
NSG	Stateful allow/deny rules — applied to subnet or individual NIC
Azure Firewall	Managed FQDN firewall — centralized inspection, threat intel
VNet Peering	Direct backbone link between VNets — same or cross-region
Azure Bastion	Browser-based RDP/SSH — no public IP needed on VMs

Hybrid Connectivity

- VPN Gateway: encrypted IPsec tunnel over public internet — up to 10 Gbps, ~\$30-500/month
- Site-to-Site (S2S): permanent on-prem to Azure VNet connection
- Point-to-Site (P2S): individual remote worker to Azure VNet
- ExpressRoute: private dedicated circuit via connectivity provider — up to 100 Gbps, NOT over internet
- ExpressRoute: lower latency, higher reliability, higher cost — required for compliance requiring private connectivity

VPN Gateway S2S	On-premises network ↔ Azure VNet — always-on encrypted tunnel
VPN Gateway P2S	Individual device ↔ Azure VNet — for remote workers
ExpressRoute	Private circuit via partner — bypasses internet, 10-100 Gbps
Virtual WAN	Managed hub connecting branches, VNets, ExpressRoute, VPN
Azure DNS	Host DNS zones in Azure — global, integrated with Azure AD

Load Balancing & Traffic Management

- Azure Load Balancer: Layer 4 TCP/UDP — distributes traffic to backend VMs, health probes
- Application Gateway: Layer 7 HTTP/HTTPS — URL routing, SSL termination, WAF protection
- Azure Front Door: global Layer 7 CDN+LB+WAF — accelerates apps worldwide using Microsoft network

- Traffic Manager: DNS-based global routing — performance, priority, weighted, geographic methods
- Private Endpoint: private IP in your VNet for Azure PaaS — traffic never leaves Microsoft backbone

Load Balancer	Layer 4 – distributes TCP/UDP traffic to VMs in backend pool
App Gateway	Layer 7 – URL-based routing, SSL offload, WAF for web apps
Azure Front Door	Global CDN+LB+WAF – accelerate and protect web applications
Traffic Manager	DNS-based global routing – choose closest/fastest endpoint
Private Endpoint	Private IP for PaaS service – no public internet exposure

Section 6: Identity, Access & Security

Entra ID, MFA, RBAC, Zero Trust, Defender & Compliance

Microsoft Entra ID (Azure AD)

- Entra ID (formerly Azure AD): cloud identity and access management — NOT the same as on-prem Active Directory
- On-prem AD uses LDAP/Kerberos; Entra ID uses OAuth 2.0, OpenID Connect, SAML (HTTP-based protocols)
- Azure AD Connect: syncs on-prem identities to Entra ID — enables hybrid identity (same credentials)
- SSO (Single Sign-On): log in once → access all cloud apps (Microsoft 365, Salesforce, 1000s more)
- MFA: requires 2+ factors — password + Authenticator app/SMS/FIDO2 key — blocks 99.9% of account attacks

Entra ID Free	Basic SSO, MFA for admins, basic reports — included with Azure
Entra ID P1	Conditional Access, hybrid identity, dynamic groups, self-service
Entra ID P2	Identity Protection (risk-based CA), PIM, access reviews
MFA Methods	Authenticator app SMS Phone call FIDO2 security key
Security Defaults	Free MFA for all users — enable if not using Conditional Access

RBAC & Zero Trust

- RBAC (Role-Based Access Control): who + what + where — Security Principal + Role + Scope
- Built-in roles: Owner (full access + assign roles), Contributor (full except assign roles), Reader (view only)
- Least Privilege: grant MINIMUM permissions required — use Reader before Contributor, scope to resource not sub
- Zero Trust: never trust, always verify — assume breach, verify explicitly, use least privilege
- Conditional Access: if (user + location + device + app + risk) → then (allow / deny / require MFA)

Owner	Full access + assign roles — use sparingly
Contributor	Full resource management — cannot assign roles or manage access
Reader	View only — use for auditors and monitoring
Conditional Access	Policy engine: signals → conditions → access decision
PIM	Just-in-time privileged access — request elevation for limited time

Azure Security Services

- Azure Key Vault: securely store secrets, keys, certificates — access via API with Azure AD auth, full audit log
- Managed Identity: auto-managed Azure AD identity for resources — no credentials in code (system-assigned or user-assigned)

- Microsoft Defender for Cloud: Secure Score + recommendations + threat detection across all Azure services
- Microsoft Sentinel: cloud-native SIEM + SOAR – AI threat detection, incident investigation, automated response
- DDoS Protection Standard: adaptive tuning, attack analytics, cost protection, 24/7 response team

Key Vault	<i>Store secrets/keys/certs – apps fetch via API, no hardcoding</i>
Managed Identity	<i>App authenticates to Azure services without any credentials/secrets</i>
Defender for Cloud	<i>Secure Score + recommendations + threat detection + compliance</i>
Microsoft Sentinel	<i>SIEM+SOAR – collect, detect, investigate, respond to threats</i>
JIT VM Access	<i>Lock RDP/SSH by default – open only on request, for limited time</i>

Section 7: Azure Management & Governance

Policy, Cost Management, Monitor, Advisor & Compliance Tools

Azure Governance Tools

- Azure Policy: enforce rules on resources — Deny (block), Audit (log), DeployIfNotExists (auto-remediate)
- Initiatives: group of policies — e.g., 'ISO 27001' initiative bundles 20+ related policies
- Azure Blueprints: package ARM templates + Policies + RBAC + RGs for repeatable compliant environments
- Azure Locks: Delete lock (prevent deletion) | Read-only lock (prevent changes AND deletion)
- RBAC vs Policy: RBAC controls WHO can act — Policy controls WHAT resources can look like

Azure Policy	Enforce rules: allowed regions, required tags, approved VM SKUs
Policy Initiative	Bundle of related policies — apply compliance framework at once
Azure Blueprints	Deploy compliant environment: RG + Policy + RBAC + ARM template
Delete Lock	Prevent accidental resource deletion — Owner can add/remove
Read-only Lock	Prevent modifications AND deletions — strongest lock

Cost Management & Pricing

- Pricing Calculator: estimate Azure costs BEFORE deployment — configure services and get monthly estimate
- TCO Calculator: compare on-premises vs Azure — estimate 3-5 year savings (typically 30-90% savings)
- Azure Cost Management: monitor ACTUAL spend — cost analysis, budgets, alerts, recommendations
- Budgets: set spending thresholds — alerts via email/webhook, does NOT stop spending automatically
- Cost factors: resource type, consumption, region, billing zone (data transfer), reserved vs PAYG

Pricing Calculator	azure.microsoft.com/pricing/calculator — estimate before deploying
TCO Calculator	On-prem vs Azure cost comparison — build migration business case
Cost Management	Monitor actual spending — dashboards, budgets, cost alerts
Azure Advisor	Free recommendations: Cost Security Reliability Performance Ops
Cost saving tips	Reserved Instances + Spot VMs + Hybrid Benefit + right-sizing

Monitoring & Management Tools

- Azure Monitor: collect metrics + logs from all Azure resources — alerts, dashboards, autoscale
- Log Analytics: query logs using KQL (Kusto Query Language) — correlate data from multiple sources
- Application Insights: APM for web apps — request rates, failures, dependencies, user behavior

- Azure Advisor: personalized recommendations across Cost, Security, Reliability, Performance, Ops Excellence
- Azure Service Health: personalized view of outages + planned maintenance affecting YOUR subscriptions

Azure Monitor	Central monitoring – metrics, logs, alerts, autoscale, dashboards
Log Analytics	Query logs with KQL – correlate Azure + on-prem + custom sources
App Insights	Web app APM – errors, latency, user sessions, deployment annotations
Azure Advisor	Actionable recommendations aligned to Well-Architected Framework pillars
Service Health	Azure Status (global) + Service Health (your subs) + Resource Health

Management Interfaces

- Azure Portal: web GUI at portal.azure.com – good for exploration, manual operations, dashboards
- Azure CLI: cross-platform command line – 'az' commands, scriptable, available on Win/Mac/Linux
- Azure PowerShell: PowerShell module – 'New-Az*', 'Get-Az*' cmdlets, preferred by Windows admins
- Azure Cloud Shell: browser-based shell in Portal – pre-authenticated CLI + PowerShell + all tools
- ARM Templates / Bicep / Terraform: Infrastructure as Code – repeatable, version-controlled deployments

Azure Portal	portal.azure.com – GUI for managing all Azure resources
Azure CLI	az login □ az vm create □ az group list (cross-platform)
Azure PowerShell	Connect-AzAccount □ New-AzVM □ Get-AzResourceGroup
Cloud Shell	Browser-based shell – pre-authenticated, all tools pre-installed
Mobile App	iOS/Android – monitor resources, view alerts, run Cloud Shell

Section 8: Azure Databases & AI

SQL, Cosmos DB, MySQL, AI Services & Azure OpenAI

Azure Database Services

- Azure SQL Database: fully managed SQL Server PaaS — auto-patching, backups, HA, serverless option
- SQL Managed Instance: near 100% SQL Server compatibility — for lift-and-shift migrations needing SQL Agent/CLR
- Azure Database for MySQL / PostgreSQL: fully managed open-source relational databases
- Azure Cosmos DB: globally distributed NoSQL — millisecond latency, multiple APIs (SQL, MongoDB, Cassandra, Gremlin, Table)
- Azure Cache for Redis: in-memory cache — microsecond latency, dramatically improves app performance

SQL Database	<i>PaaS SQL Server — serverless option pauses when idle (saves cost)</i>
SQL Managed Instance	<i>Full SQL Server engine in cloud — SQL Agent, CLR, linked servers</i>
Cosmos DB	<i>Multi-model NoSQL — global distribution, 5 consistency levels, 99.999% SLA</i>
Cosmos DB RU	<i>Request Unit = currency for throughput — 1 RU = read 1 KB item</i>
Redis Cache	<i>In-memory key-value store — cache DB results for microsecond access</i>

Azure AI & Cognitive Services

- Azure AI Services: pre-built AI APIs — Vision, Language, Speech, Decision — no ML expertise needed
- Azure Machine Learning: full ML platform — AutoML, Designer, Notebooks, model training and deployment
- Azure OpenAI: GPT-4, DALL-E, Codex models with enterprise security — VNet, Azure AD, content filtering
- Azure Bot Service: build chatbots connecting to Teams, Slack, SMS, web — uses AI Language for NLU
- Responsible AI principles: Fairness | Reliability & Safety | Privacy & Security | Inclusiveness | Transparency | Accountability

Computer Vision	<i>Detect objects, OCR, spatial analysis, custom vision models</i>
AI Language	<i>Sentiment, NER, translation, Q&A, intent classification (CLU)</i>
Azure OpenAI	<i>GPT-4 / DALL-E / Codex with enterprise security and compliance</i>
Azure ML AutoML	<i>Automated ML — finds best algorithm for your data, no code needed</i>
Document Intelligence	<i>Extract structured data from invoices, receipts, forms using AI</i>

Analytics & Integration Services

- Azure Synapse Analytics: unified analytics — SQL pools (data warehouse) + Spark pools (big data) + Pipelines
- Azure Data Factory: ETL/ELT service — ingest from 90+ sources, transform, load to Azure analytics
- Azure Event Hubs: big data streaming — millions of events/second, Kafka-compatible, telemetry ingestion
- Azure IoT Hub: bidirectional IoT communication — device management, telemetry, cloud-to-device commands
- Power BI: business intelligence — connects to Azure SQL, Synapse, Blob; embed dashboards in apps

Synapse Analytics	<i>Unified data warehouse + big data — SQL + Spark in one service</i>
Azure Data Factory	<i>Cloud ETL — move data from 90+ sources to Azure analytics</i>
Event Hubs	<i>High-throughput streaming — IoT telemetry, clickstreams, log data</i>
Stream Analytics	<i>Real-time SQL-like queries on streaming data — millisecond latency</i>
IoT Central	<i>Managed IoT SaaS — connect devices, dashboards, rules — no code</i>

Section 9: Azure DevOps & Developer Tools

DevOps Services, CI/CD, IaC & Modern Development

Azure DevOps Services

- Azure Boards: Agile project management — backlogs, sprints, Kanban, Scrum, work item tracking
- Azure Repos: unlimited free Git repositories — branch policies, pull requests, code review
- Azure Pipelines: CI/CD for any language, any platform — YAML-defined, hosted or self-hosted agents
- Azure Artifacts: private package feeds — NuGet, npm, Maven, Python, Universal Packages
- Azure Test Plans: manual and exploratory testing, test case management, browser-based test runner

Azure Boards	Agile planning — Epics □ Features □ User Stories □ Tasks hierarchy
Azure Repos	Git hosting — branch protection, PR policies, code search
Azure Pipelines	CI: build+test on every commit CD: auto-deploy to environments
GitHub Actions	Native CI/CD for GitHub repos — large marketplace of actions
Azure Artifacts	Private NuGet/npm/Maven/Python feeds + upstream public proxying

Infrastructure as Code & Tools

- ARM Templates: JSON declarative IaC — idempotent, version-controllable, all Azure resources
- Bicep: cleaner DSL compiling to ARM — preferred for Azure-native IaC, VS Code IntelliSense support
- Terraform: multi-cloud IaC (Azure+AWS+GCP) — state file tracks deployed resources
- Azure CLI + PowerShell: imperative scripting — automate resource management in CI/CD pipelines
- Azure Static Web Apps: auto-build + deploy from GitHub/DevOps — global CDN + serverless API backend

Bicep	<code>az bicep build file.bicep</code> □ ARM JSON □ <code>az deployment group create</code>
Terraform	<code>terraform init</code> □ <code>plan</code> □ <code>apply</code> — state stored in Azure Storage
ARM What-if	Preview changes before deploying ARM template — like Terraform <code>plan</code>
Azure Static Web Apps	JAMstack hosting: GitHub commit □ build □ CDN deploy automatically
Azure App Config	Centralized feature flags and config — no redeployment for changes

Messaging & Integration

- Azure Service Bus: enterprise message broker — queues (FIFO) + topics (pub/sub), transactions, dead-letter
- Azure Event Grid: serverless event routing — publish-subscribe, fan-out to Functions/Logic Apps/webhooks
- Azure API Management: API gateway — rate limiting, OAuth, caching, developer portal, analytics

- Azure SignalR: managed real-time WebSocket connections — live dashboards, chat, multiplayer games
- Event Grid vs Service Bus vs Event Hubs: routing vs messaging vs streaming — choose based on pattern

Service Bus Queue	One-to-one reliable messaging — FIFO, at-least-once, transactions
Service Bus Topic	One-to-many pub/sub — filtered subscriptions per consumer
Event Grid	Event routing (something happened) □ trigger Functions/Logic Apps
Event Hubs	High-throughput streaming — big data ingestion pipeline
API Management	Gateway: auth, rate limit, transform, monitor APIs centrally

Section 10: Pricing, Support & Exam Tips

SLAs, Support Plans, Compliance & AZ-900 Exam Strategy

Azure SLAs & Pricing Models

- SLA: uptime guarantee — 99.9% = 8.7 hrs/year downtime | 99.99% = 52 min/year | 99.999% = 5.3 min/year
- Composite SLA: multiply individual SLAs — adding dependencies LOWERS the overall SLA
- No SLA for free tier services or services in preview — always check SLA for production workloads
- Pricing factors: resource type, region, usage (meters), bandwidth (data out), reserved vs PAYG
- Free services always: Azure AD free, Functions (1M exec), Blob (5 GB), Cosmos (1,000 RU/s), DevOps (5 users)

Pricing Calculator	Estimate costs BEFORE deploying — configure + get monthly estimate
TCO Calculator	Compare on-prem vs Azure — show savings to justify migration
PAYG	Pay-as-you-go — no commitment, highest flexibility, highest per-unit cost
Reserved Instances	1 or 3 year commitment — up to 72% savings, cancel with fee
Spot VMs	Use unused capacity — up to 90% saving, evicted with 30-sec notice

Azure Support Plans

- Basic (free): documentation, self-help, Azure Advisor, Service Health — NO technical support
- Developer (\$29/mo): business hours email support, 8-hr response for Sev C
- Standard (\$100/mo): 24/7 phone+email, 2-hr Sev B, 4-hr Sev C response
- Professional Direct (\$1,000/mo): 1-hr Sev A, proactive guidance, architecture reviews
- Premier/Unified: enterprise — dedicated Technical Account Manager (TAM), custom SLA, mission-critical

Basic	Free — self-help only, no ticket support, Advisor + Service Health
Developer	\$29/mo — business hours email, good for non-production/testing
Standard	\$100/mo — 24/7 support, min recommended for production workloads
Professional Direct	\$1,000/mo — fastest response, proactive guidance for enterprises
Severity A	Critical business impact — 1-hr response (Standard: not included)

Compliance & Trust

- Azure has 100+ compliance certifications: ISO 27001, SOC 1/2/3, PCI DSS L1, HIPAA, FedRAMP, GDPR
- Service Trust Portal (servicetrust.microsoft.com): download audit reports, certifications, compliance guides
- Microsoft Privacy Statement: Microsoft is a data processor — customer owns data, used only to provide service

- Customer Lockbox: approve/deny Microsoft engineer access to your data during support cases
- Azure Sovereign Clouds: Azure Government (US), Azure China (21Vianet), Azure Germany – isolated instances

Service Trust Portal	<i>servicetrust.microsoft.com – audit reports and certifications</i>
Microsoft Purview	<i>Unified data governance – classify, catalog, protect data at scale</i>
Azure Policy	<i>Enforce compliance rules across all resources in subscriptions</i>
Customer Lockbox	<i>You approve/deny MS engineer data access – logged for audit</i>
Compliance Manager	<i>Track your organization's compliance against frameworks</i>

AZ-900 Exam Quick Reference

- Exam format: 40-60 questions | 65 minutes | Passing score: 700/1000 | Cost: \$165 USD
- Skill areas: Cloud Concepts (25-30%) | Azure Architecture+Services (35-40%) | Management+Governance (30-35%)
- Key concepts to master: IaaS/PaaS/SaaS | Shared Responsibility | CapEx/OpEx | Availability Zones vs Sets
- Know the tools: Pricing Calc vs TCO Calc | Azure Advisor | Azure Monitor | Service Health | Cost Management
- AZ-900 does NOT test: implementation details, CLI commands, ARM template syntax – only CONCEPTS

IaaS/PaaS/SaaS	<i>Know which Azure services fall in each category – exam favourite</i>
Shared Responsibility	<i>Customer vs Microsoft responsibility shifts IaaS↔PaaS↔SaaS</i>
Availability Zones	<i>3 separate DCs per region – 99.99% SLA – protects DC failure</i>
Availability Sets	<i>Fault + update domains in ONE DC – 99.95% SLA – exam trap!</i>
Microsoft Learn	<i>learn.microsoft.com/certifications/azure-fundamentals – free, official</i>